

Salifort Motors

Employee Retention Project

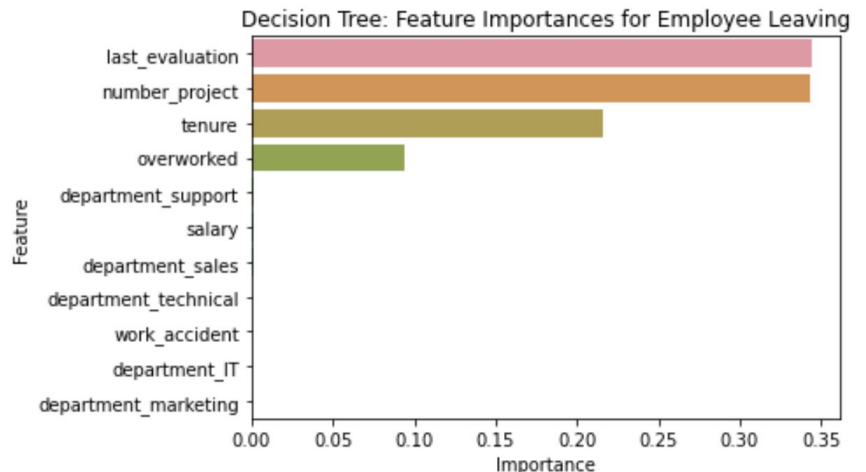
Project Overview

The HR department at Salifort Motors wants to take some initiatives to improve employee satisfaction levels at the company. They collected data from employees, but now they don't know what to do with it. They refer to you as a data analytics professional and ask you to provide data-driven suggestions based on your understanding of the data. They have the following question: what's likely to make the employee leave the company?

Details

Key Insights

- **Models are highly effective** at predicting attrition, with **96.5% accuracy**.
- **Last Evaluation is the top predictor**, indicating that both **underperformers and top performers are at risk of leaving**.
- **Workload imbalance** is a major driver, as employees with either **too few or too many projects and hours** are more likely to quit.
- **Time spent at the company** is a critical factor, with attrition risk peaking at **4-5 years of tenure**, suggesting a need for a clear career path.
- **Low employee satisfaction** is a foundational driver of attrition, reinforcing the importance of addressing the above issues.



Next Steps

- **Implement Recommendations:** Address workload imbalance and low satisfaction through targeted initiatives.
- **Develop Career Paths:** Create structured promotion plans to retain high performers and long-tenured employees.
- **Deploy Predictive Model:** Integrate the predictive model into HR workflows to proactively identify and support at-risk employees.

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ISSUE / PROBLEM

Salifort Motors seeks to improve employee retention and answer the following question:

What's likely to make the employee leave the company?

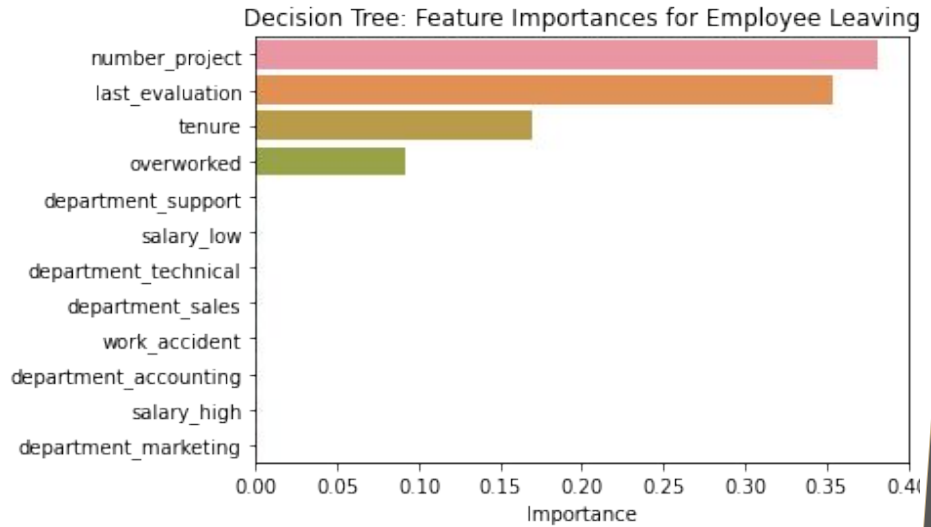
RESPONSE

Since the variable we are seeking to predict is categorical, the team could build either a logistic regression or a tree-based machine learning model.

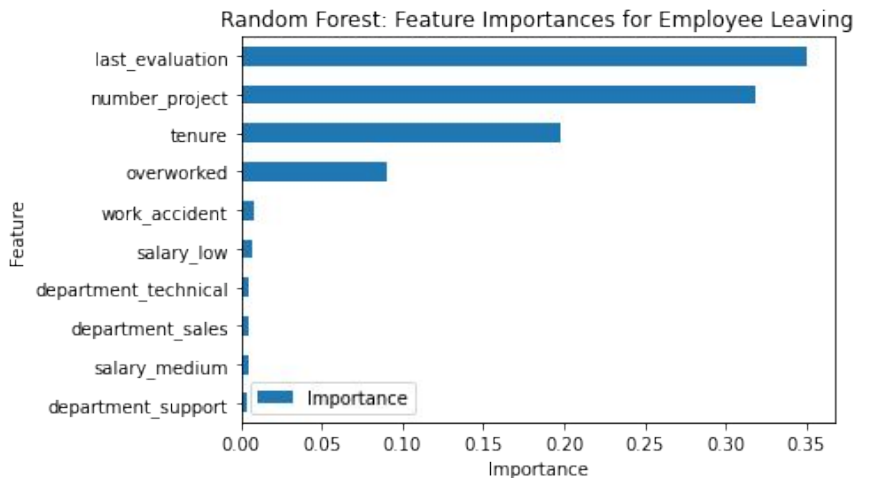
The random forest model slightly outperforms the decision tree model.

IMPACT

This model helps predict whether an employee will leave and identify which factors are most influential. These insights can help HR make decisions to improve employee retention.



Barplot above shows the most relevant variables: **'last_evaluation'**, **'number_project'**, **'tenure'** and **'overworked'**.



In the random forest model above, **'last_evaluation'**, **'tenure'**, **'number_project'**, **'overworked'**, **'salary_low'**, and **'work_accident'** have the highest importance. These variables are most helpful in predicting the outcome variable, **'left'**.

KEY INSIGHTS

- Cap the number of projects that employees can work on.
- Consider promoting employees who have been with the company for at least four years, or conduct further investigation about why four-year tenured employees are so dissatisfied.
- Either reward employees for working longer hours, or don't require them to do so.
- If employees aren't familiar with the company's overtime pay policies, inform them about this. If the expectations around workload and time off aren't explicit, make them clear.
- Hold company-wide and within-team discussions to understand and address the company work culture, across the board and in specific contexts.
- High evaluation scores should not be reserved for employees who work 200+ hours per month. Consider a proportionate scale for rewarding employees who contribute more/put in more effort.